Project Documentation

Student Exam Performance Dataset Analysis (Python Project)

1.Project Overview

This dataset shows the student-related academic, lifestyle, and socio-enconmic factores that include exam performance. It include study hours,student attendence,sleep hours, motivation,tutoring sessions,parental involvement,school type and other envirmental features.The main target varibale is Exam_Score which represents the student final exam score.This project analyses data using python to evaluate how various factors influence in scoring in exam.

2.Tools used

- Python
- Pandas
- Numpy
- Matplotlib
- Seaborn
- Jupyter Notebook

3.Dataset

Source: https://www.kaggle.com/datasets/grandmaster07/student-exam-performance-dataset-analysis/data

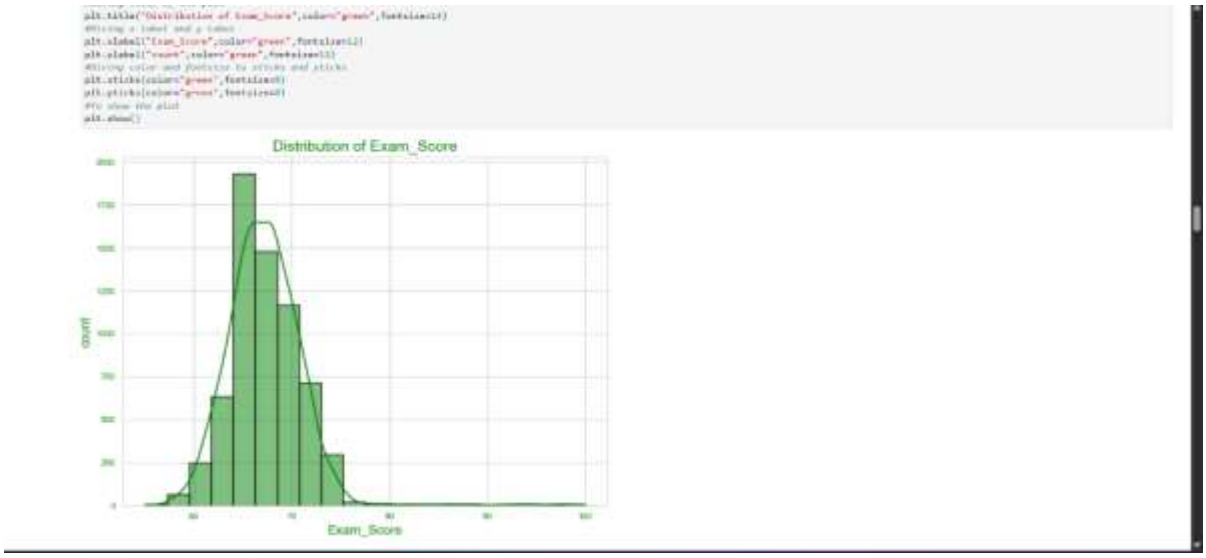
The attributes to be analysed are Hours_Studied, Attendance, Parental_Involvement Access_to_Resources, Extracurricular_Activities, Sleep_Hours,Previous_Scores, Motivation_Level,Internet_Access,Tutoring_Sessions,Family_Income,Teacher_Quality, School_Type,Peer_Influence,Physical_Activity, Learning_Disabilities, Parental_Education_Level, Distance_from_Home,Gender and Exam_Score

4.Steps Followed

I. Data Loading and Initial Overview
II. Data understanding and Data Pre-processing
III. Exploratory Data Analysis (EDA)
IV. Visualization

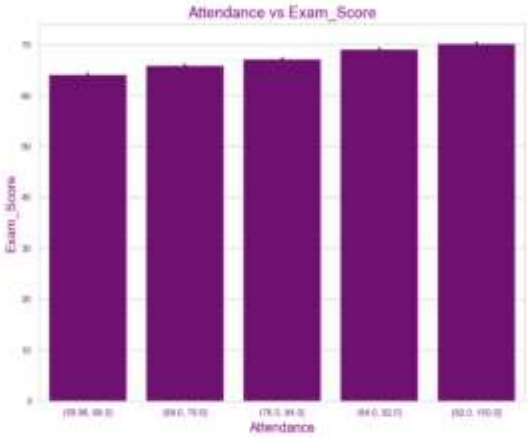
Performed univariate,bivariate and multivariate visual analyses such as Scatter plot,Bar chart,Box plot,Correalation Heat map,count plot,line graph,pair plot,pie chart to understand how various factors influences performance

5.Screenshots



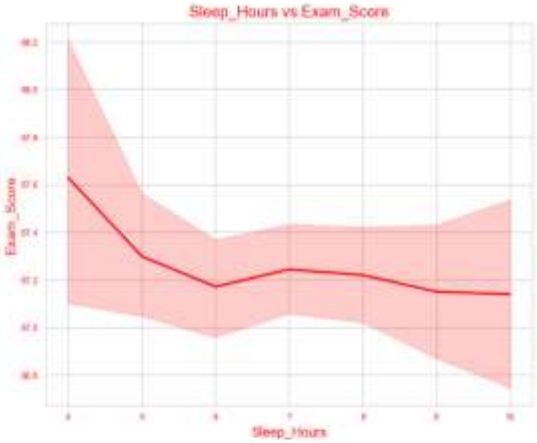
The Scatter plot Hours_Studied vs Exam_Score indicates that the female students who scored high marks studied 10-20 hours whereas the male students with high score studied 20-30 hours.This suggest that female students grasp faster than male students.It highlights the difference in the study efficiency among genders in the given dataset.

```
plt.show()
```



The Barplot indicates that the students with maximum attendance scored high marks,which implies attendance is an influencing factor in scoring high marks

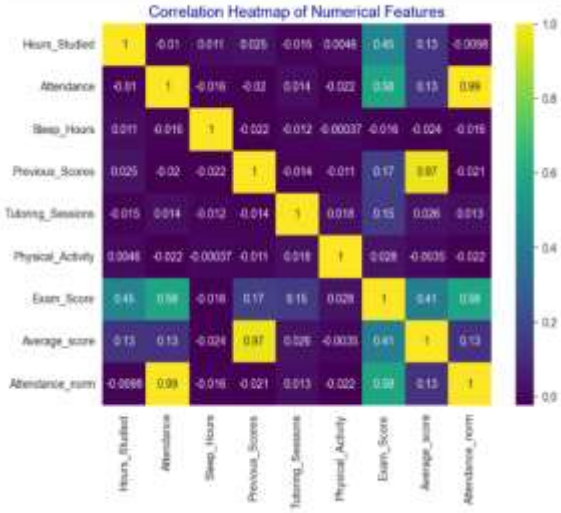
```
In [20]: #Visualization 3
#line plot
plt.figure(figsize=(8,6))
sns.lineplot(x="Sleep_Hours",y="Exam_Score",dataset,color="red",linewidth=2)
plt.title("Sleep_Hours vs Exam_Score",color="red",fontsize=14)
```



The above linegraph portrays that the students sleep more scored less marks,instead those who sleep less have scored more.Sleep_hour is also an affecting factor in scoring marks

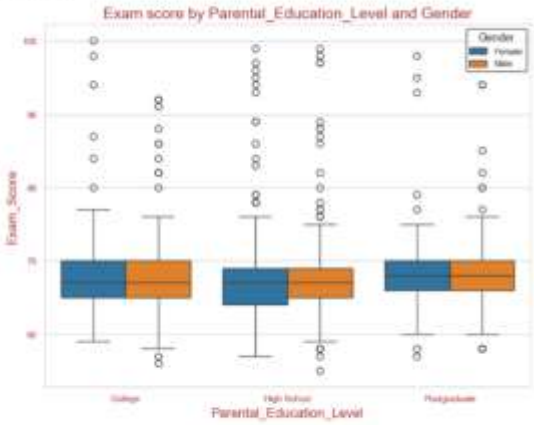
```
In [20]: #Visualization 4
#Multivariate analysis
#Heatmap
plt.figure(figsize=(8,6))
#calculating correlation between numerical columns ,showing values in each column and choosing the color style
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='viridis',linecolor='black')
```

```
plt.title("Correlation Heatmap of Numerical Features",color="darkblue",fontsize=14)
plt.show()
```



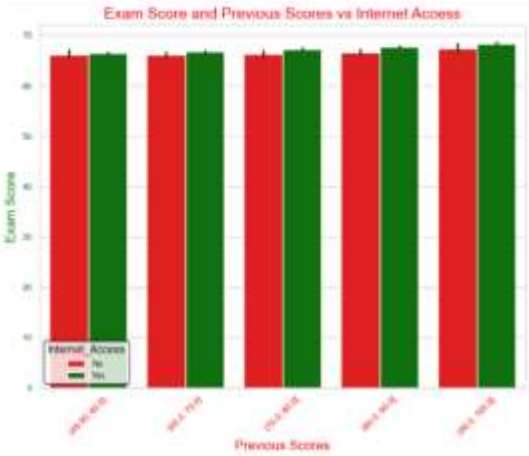
The correlation heatmap indicates how the numerical values are related .The value ranges from -1 to 1,1 indicates both increases together and -1 indicates inverse relationship and 0 means no linear relationship

```
plt.legend(title="Gender",title_fontsize=10, fontsize=8,loc="upper right",frameon=True,edgecolor="black")
plt.show()
```



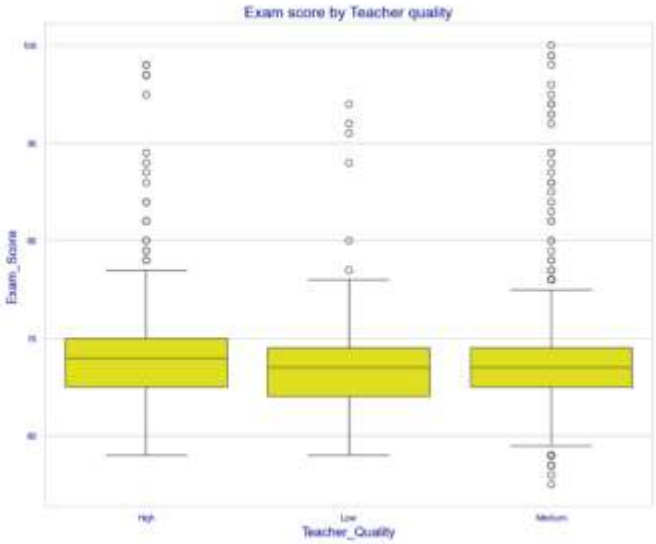
The boxplot Exam score by Parental_Education_Level and Gender reveals that in all parental_Education_Levels female students scored high marks and also most students have high scores with High school Parental_Education_Level, which implies the Parental_Education_Level doesn't influence to score more.

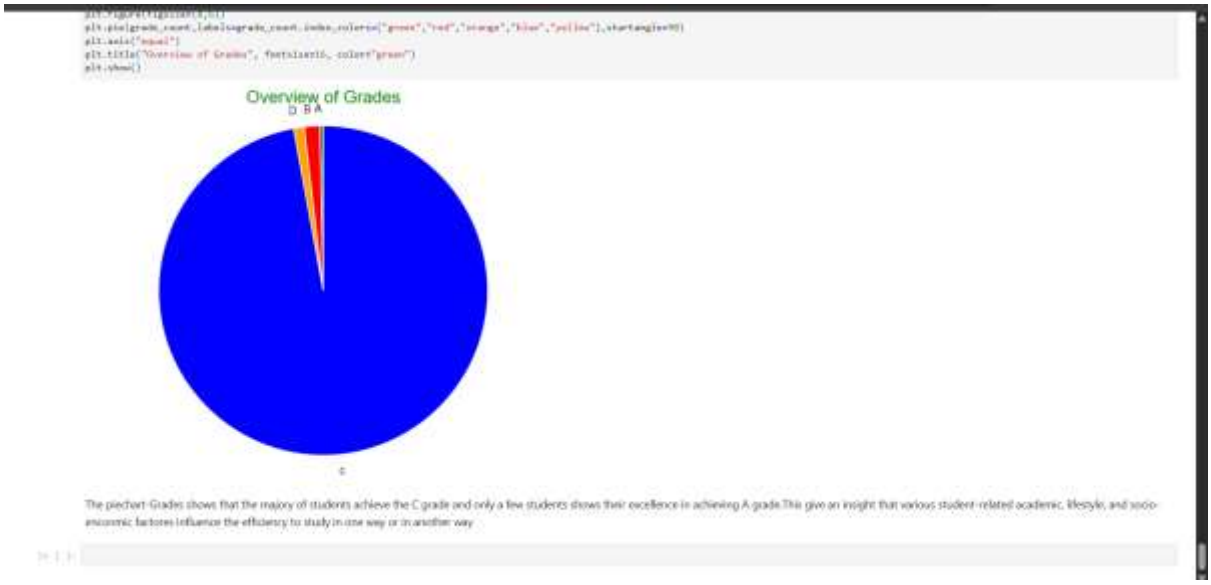
```
#Visualization 8
#Barplot
plt.figure(figsize=(10,8))
plt.show()
```



The Bargraph Exam Score and Previous Scores vs Internet Access Shows that the students with Internet access have scored high marks than others.That implies accessing internet for studies is helpful for scoring marks according to this dataset.

```
#Visualization 10
#Boxplot
plt.show()
```





5.Key Insights

- The most of the students scored in between 60 and 70 marks .Eventhough male students are more numerous than females students ,female students scored highest maks than male students
- The female students who scored high marks studied 10-20 hours whereas the male students with high scores studied 20-30 hours.This sugest that female students grasp faster than male students.It highlights the differrence in the study efficiency among genders in the given dataset.
- The students with maximum attendance and students with Internet access have scored high marks than others
- The students sleep more scored less marks,instead those who sleep less have scored more.Sleep_hour is also an affecting facor in scoring marks
- In all parental_Education_Levels female students scored high marks and also most students have high scores with High school Parental_Education_Level, which implies the Parental_Education_Level dosen't influence the score more.
- The teachers with medium quality helps students to score high and the low quality teachers are not much beneficial for students to score high marks.At the same time high quality teachers influences in a moderate way

5.Recommendations

In the given dataset majority of students achieve C grade and only a few students shows their excellence in achieving A grade.This give an insight that various student-related academic, lifestyle, and socio-enconmic factores influence the efficiency to study in one way or in another way.But elevating the usage of internet for studies, quality of teachers, study time (without effecting sleep) will be helpful for getting high exam score

6. Files Included

- Dataset

https://drive.google.com/file/d/1PlaweD9RUABcojO6L0bTsL45jdY2pFz5/view?usp=sharing

- Python Script

https://drive.google.com/file/d/17WlwzfGwQL-pasBd8egFWnfcI4CMDMEg/view?usp=drive_link

- Project HTML

https://drive.google.com/file/d/1OfCt9LkPG2QM_qGlw00pjnZLh4NbmuQr/view?usp=sharing

8. How to Run the Project

Open the link to view the python script and then run the code